I. Quadrature Oscillator Design (Q2)

A quadrature oscillator is a circuit that generates two sinusoidal signals of the same frequency but with a $90°$ phase difference. These are commonly referred to as the Sine (v_{OSCQ}) and Cosine (v_{OSCI}) signals. In this project, we designed a 3-opamp oscillator to meet the target frequency requirements using UA741 op-amps.

A. Circuit Topology and Operation

Our design, as shown in the LTSpice schematic (Fig. 1), utilizes a loop consisting of two integrators and a feedback/limiter stage.

1) **Integrator Stages (U1 and U2):** Each integrator provides a $90°$ phase shift. By cascading two of them, we get a total of $180°$ shift.
2) **The Limiter Stage (U3):** A major problem with oscillators is that the amplitude can grow until it clips against the power rails, turning the sine wave into a square wave. To prevent this, we used a third op-amp stage with 1N4148 diodes in the feedback path (R_8, R_9, R_{10}). These diodes act as a "soft-clipper," stabilizing the amplitude at a level where the waveform remains sinusoidal.
3) **Barkhausen Criterion:** For the circuit to oscillate, the total phase shift around the loop must be $360°$ (or $0°$) and the loop gain must be exactly 1. The third op-amp provides the final $180°$ inversion to satisfy this condition.

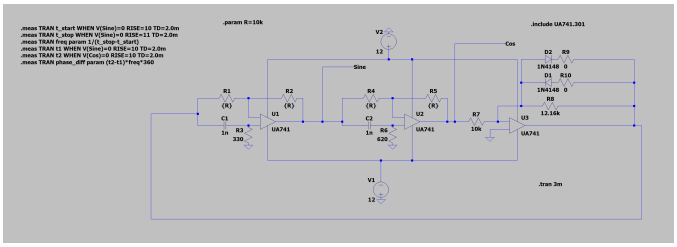


Fig. 1. LTSpice Schematic of the 3-OpAmp Quadrature Oscillator using UA741.

B. Mathematical Derivation

The transfer function of an ideal inverting integrator is given by:

$$H(s) = -\frac{1}{sRC} \tag{1}$$

In the frequency domain ($s = j\omega$):

$$H(j\omega) = -\frac{1}{j\omega RC} = \frac{j}{\omega RC} \tag{2}$$

The presence of the 'j' term indicates a $90°$ phase shift relative to the input.

If the output of the first integrator (U1) is V_{Sine}, and it is fed into the second integrator (U2), the output V_{Cos} will be:

$$V_{Cos} = V_{Sine} \times \left(\frac{1}{j\omega RC}\right) \tag{3}$$

This ensures that V_{Cos} and V_{Sine} are in quadrature. The theoretical frequency of oscillation is determined by:

$$f_{osc} = \frac{1}{2\pi RC} \tag{4}$$

Using our component values of $R = 10k\Omega$ and $C = 1nF$, the ideal frequency would be approximately $15.9kHz$. However, in our simulation, we observe a much higher frequency ($\approx 170kHz$). This shift is primarily due to the non-ideal behavior of the UA741 op-amp (specifically its limited Gain-Bandwidth Product and internal phase shifts) at high frequencies.

C. Simulation Results and Analysis

We performed a transient analysis for $3ms$ to allow the oscillations to settle. The results were captured using LTSpice's '.meas' directives and FFT analysis.

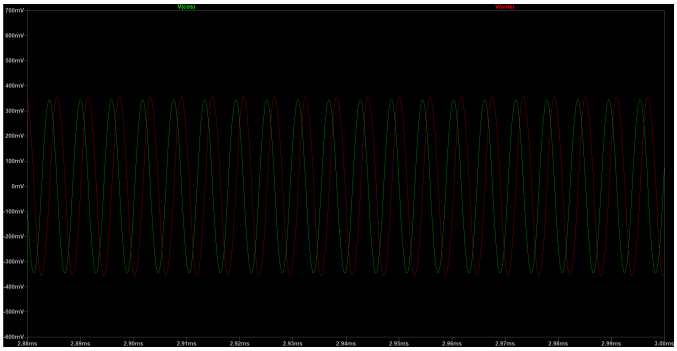


Fig. 2. Transient waveform showing the quadrature Sine (red) and Cosine (green) outputs.

1) Frequency Measurement: As shown in the Spice Error Log (Fig. 3), the measured frequency was $170.76kHz$. While this differs from the simple $1/2\pi RC$ calculation, it is stable and provides a strong "carrier" signal for our mixer stage.

2) Phase Difference: The most critical requirement is the $90°$ phase shift. Using the time-delay measurement between the zero-crossings of the two waves:

$$Phase_Diff = (t_2 - t_1) \times f \times 360° \tag{5}$$

Our simulation yielded a phase difference of $-93.38°$ (Fig. 3). This is extremely close to the ideal $90°$.

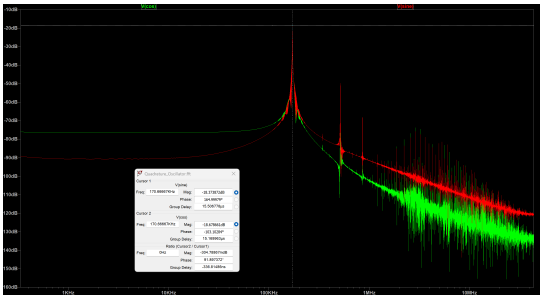


Fig. 3. LTSpice measurement log showing $f = 170.76kHz$ and Phase Difference $\approx -93.4°$.

3) Spectral Purity (FFT): The FFT plot (Fig. 4) shows a sharp peak at the fundamental frequency of $170.8kHz$. The fundamental remains dominant enough for effective mixing.

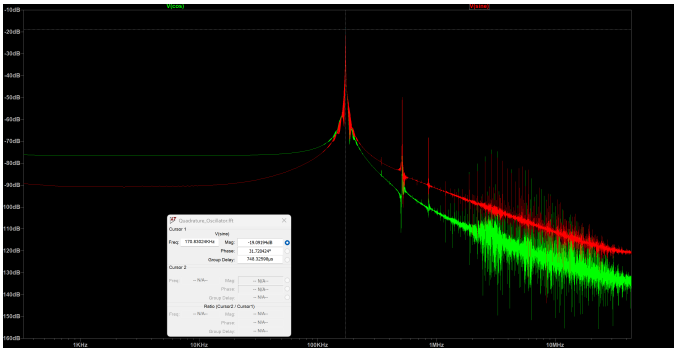


Fig. 4. FFT of the oscillator outputs showing the peak at $170.8kHz$.

D. Hardware Realization and Measurements

To validate the simulated design, the 3-opamp quadrature oscillator was physically constructed on a breadboard using discrete UA741 operational amplifiers.

1) Transient and Phase Verification: Trimpots were utilized to precisely dial in the phase and amplitude. The oscilloscope captures (Fig. 5 and Fig. 6) confirm successful oscillation and verify the target $90°$ phase separation.



Fig. 5. Oscilloscope capture showing the measured hardware oscillator frequency.



Fig. 6. Oscilloscope capture verifying the 90-degree phase separation between the physical I and Q channels.

2) Spectral Purity (Hardware FFT): FFT analysis was performed directly on the oscilloscope. As shown in Fig. 7 and Fig. 8, both signals exhibit a strong fundamental frequency peak.



Fig. 7. Hardware FFT of the Sine (In-Phase) signal.



Fig. 8. Hardware FFT of the Cosine (Quadrature) signal.

E. Discussion

The design successfully generates two quadrature signals both in software and on the physical bench. Although the UA741 has significant speed limitations, we achieved stable

oscillation in the $170kHz$ range. This oscillator is fully verified to serve as the Local Oscillator (LO) for the subsequent mixer stages.